

Overview in General

This file contains a general overview of the data in the graph including node labels and relationships types.

References

- [jqassistant](#)
- [Neo4j Python Driver](#)

Node Labels

Table 1a - Highest node count by label combination

Lists the 30 label combinations with the highest number of nodes. The labels with the lowest node count are listed in table 1b. The total list would sum up to the total number of labels (100%).

The whole table can be found in the CSV report `Node_label_combination_count`.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Git, Update, Change]	178996	49.272594
1	[Git, Change, Create]	53059	14.605659
2	[Git, Change, Delete]	28004	7.708718
3	[Java, ByteCode, Member, Method]	13451	3.702684
4	[Java, ByteCode, Parameter]	13293	3.659191
5	[Git, Commit]	13149	3.619552
6	[File, Git]	11241	3.094333
7	[Git, Change, Rename]	10854	2.987803
8	[Java, ByteCode, ParameterizedType, Bound]	7279	2.003705
9	[Java, ByteCode, Bound]	7240	1.992970
10	[Java, ByteCode, Member, Field]	3604	0.992080
11	[Java, ByteCode, Bound, WildcardType]	2947	0.811227
12	[Value, Java, ByteCode, Annotation]	2931	0.806822
13	[Xml, Element]	2162	0.595138
14	[Java, ByteCode, Member, Method, Constructor]	2109	0.580549
15	[Xml, Text]	1450	0.399144
16	[Java, ByteCode, Bound, TypeVariable]	1111	0.305827
17	[Java, ByteCode, Member, Method, Lambda]	968	0.266463
18	[Type, File, Java, ByteCode, ResolvedDuplicate...	888	0.244442
19	[Type, File, Java, ByteCode, Class]	846	0.232880
20	[Json, Key]	702	0.193241
21	[Json, Value, Scalar]	685	0.188561
22	[Value, Java, ByteCode, Primitive]	672	0.184983
23	[Type, File, Java, ByteCode, JavaType]	649	0.178652
24	[Java, ByteCode, Member, Method, GenericDeclar...	578	0.159107
25	[Type, File, Java, ByteCode, ExternalType]	411	0.113137
26	[Git, Change, Copy]	342	0.094143
27	[Author, Git, Person]	299	0.082306
28	[Value, Array]	287	0.079003
29	[Committer, Git, Person]	252	0.069369

Chart 1a - Highest node count by label combination

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in Chart 1b.

<Figure size 640x480 with 0 Axes>

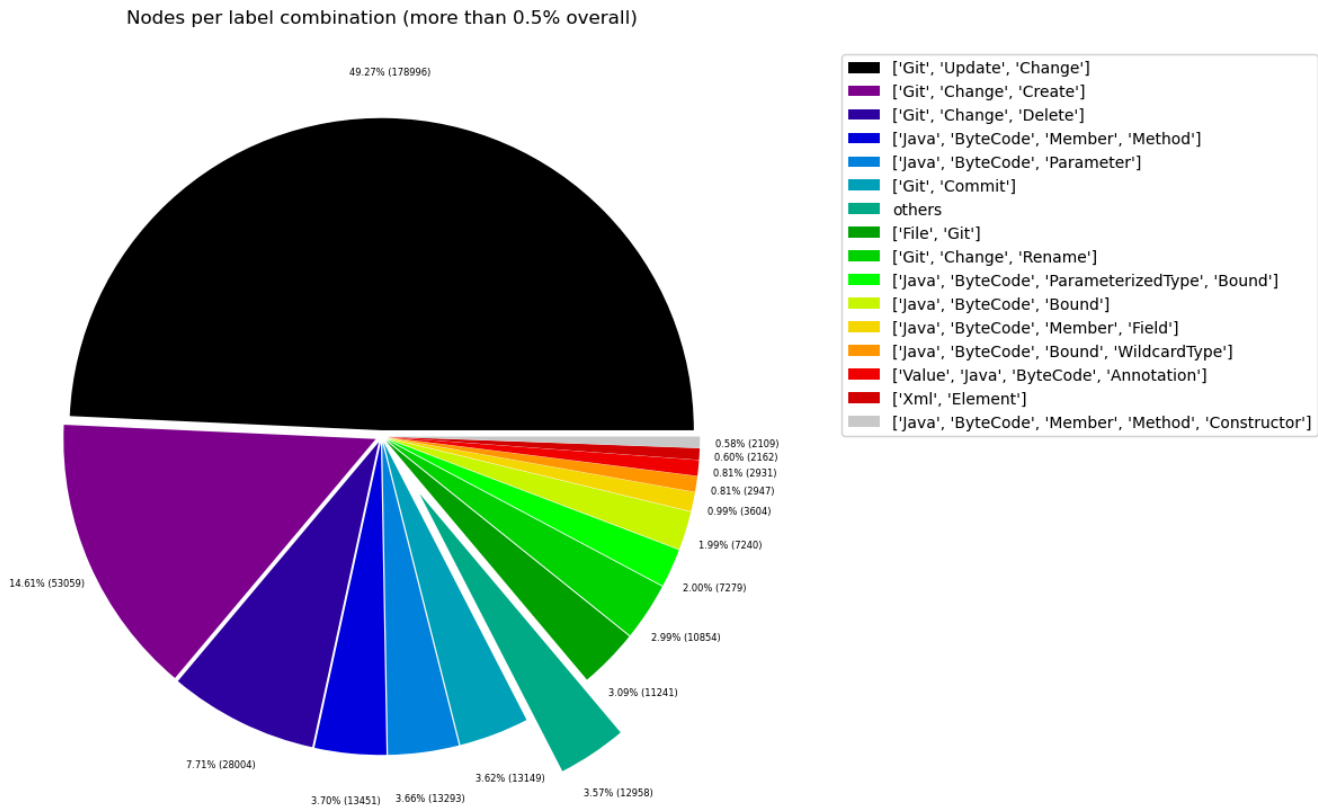


Table 1b - Lowest node count by label combination

Lists the 30 label combinations with the lowest number of nodes until they reach 0.5% of the total node count, which are shown above.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Analyze, Task, jQAssistant]	1	0.000275
1	[Package, File, Json, NPM]	1	0.000275
2	[Repository, File, Git]	1	0.000275
3	[File, TS, Scan]	1	0.000275
4	[File, Json]	2	0.000551
5	[File]	3	0.000826
6	[Java, ByteCode, Member, Method, GenericDeclar...	4	0.001101
7	[Maven, Exclusion]	5	0.001376
8	[Json, Value, Array]	6	0.001652
9	[NPM, Dependency]	7	0.001927
10	[Type, File, Java, ByteCode, Void]	9	0.002477
11	[File, Maven, Xml, Pom, Document]	9	0.002477
12	[Java, ManifestSection]	9	0.002477
13	[File, Java, Manifest]	9	0.002477
14	[File, Artifact, Jar, Archive, Zip, Java]	9	0.002477
15	[File, Java, ServiceLoader]	10	0.002753
16	[File, Java, Properties]	12	0.003303
17	[Maven, ExecutionGoal]	16	0.004404
18	[Maven, PluginExecution]	16	0.004404
19	[Xml, Attribute]	18	0.004955
20	[jQAssistant, Rule, Concept]	19	0.005230
21	[Type, File, Java, ByteCode, Throwable, Extern...	21	0.005781
22	[Maven, Configuration]	21	0.005781
23	[Maven, Plugin]	21	0.005781
24	[Type, File, Java, ByteCode, Throwable, Resolv...	22	0.006056
25	[Type, File, Java, ByteCode, Enum]	28	0.007708
26	[Git, Branch]	30	0.008258
27	[Type, File, Java, ByteCode, PrimitiveType]	30	0.008258
28	[Xml, Namespace]	36	0.009910
29	[Type, File, Java, ByteCode, Annotation]	44	0.012112

Chart 1b - Lowest node count by label combination

Shows the lowest (less than 0.5% overall) node count label combinations. Therefore, this plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

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Nodes per label combination (less than 0.5% overall)

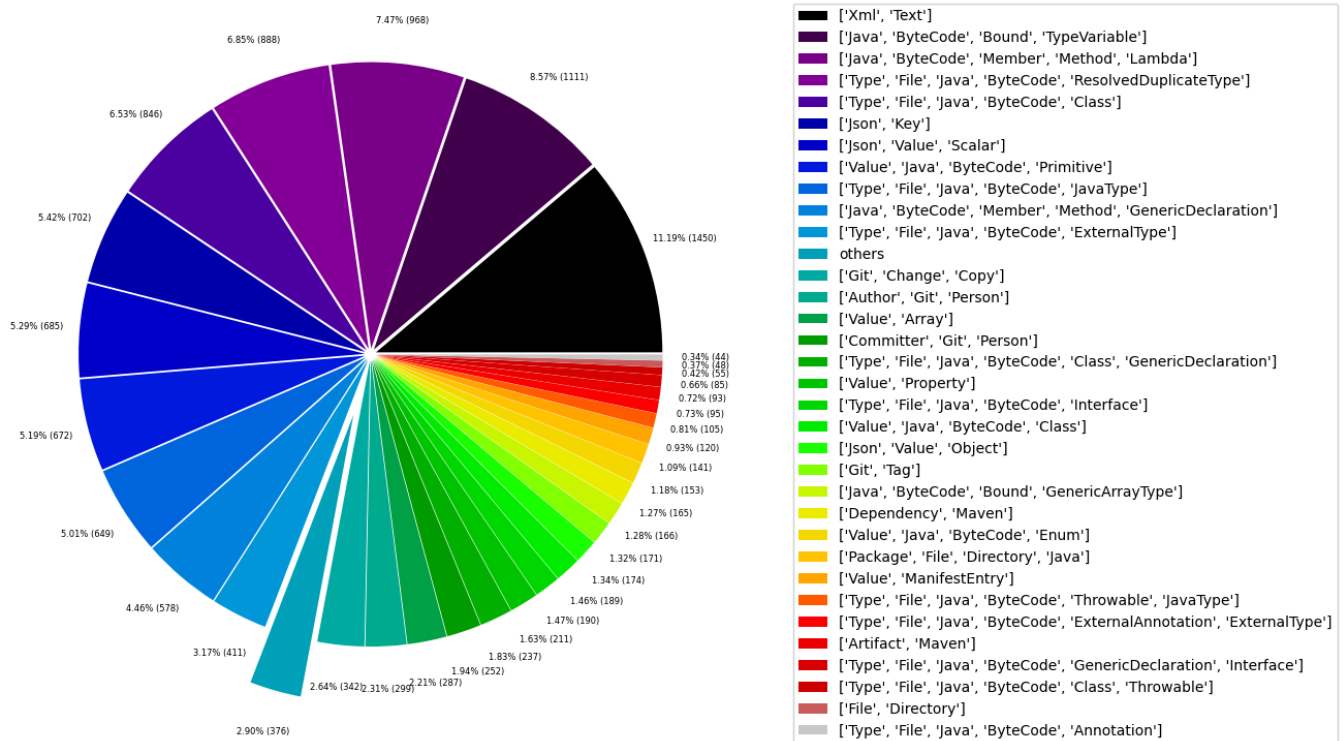


Table 1c - Highest node count by single label

Lists the 40 labels with the highest number of nodes. Doesn't sum up to the total number of nodes or 100% because one node can have multiple labels. Helps to identify commonly used labels.

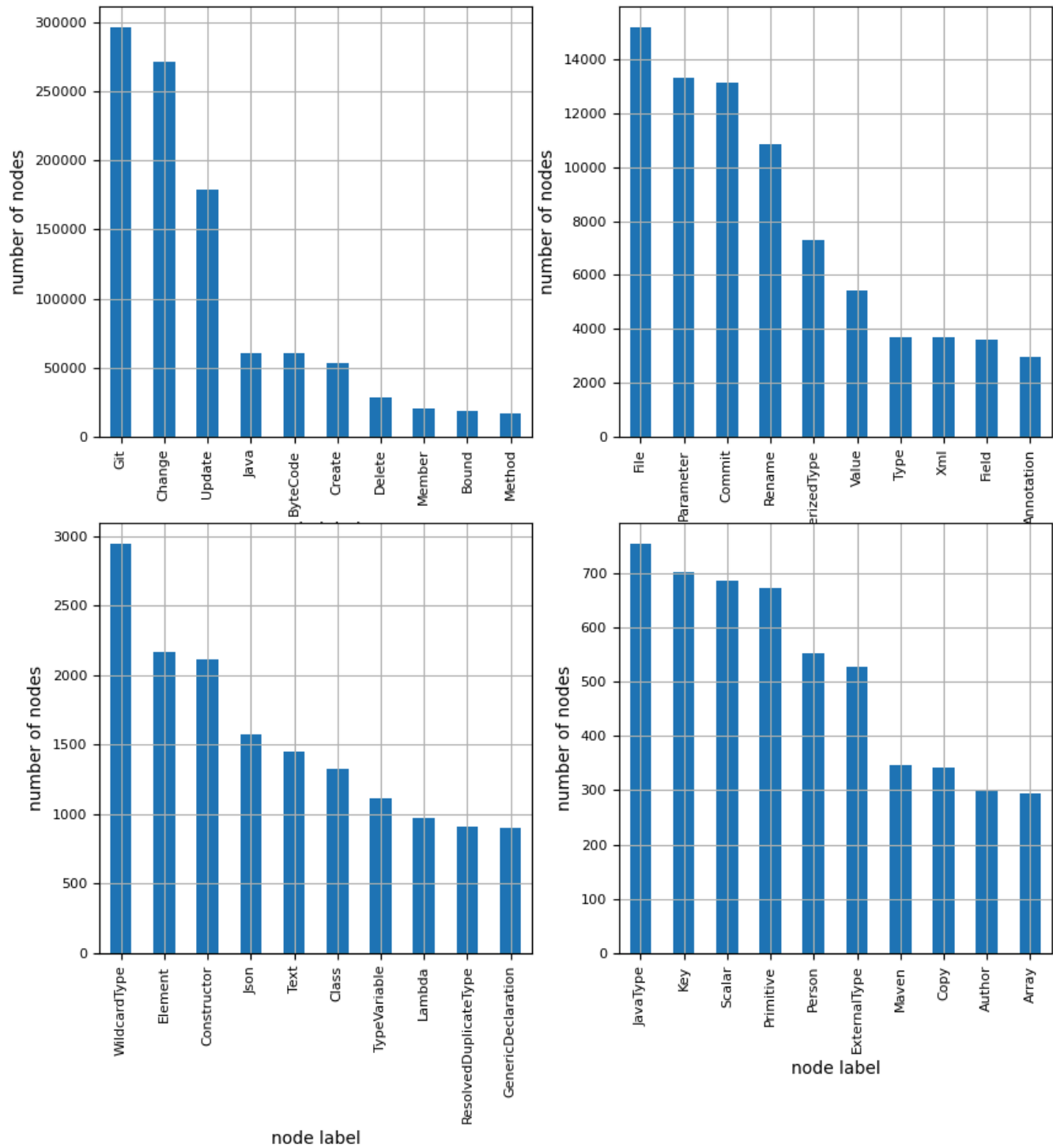
	nodeLabel	nodesWithThatLabel	nodesWithThatLabelPercent
0	Git	296398	81.590081
1	Change	271255	74.668917
2	Update	178996	49.272594
3	Java	60600	16.681485
4	ByteCode	60410	16.629184
5	Create	53059	14.605659
6	Delete	28004	7.708718
7	Member	20714	5.701985
8	Bound	18743	5.159424
9	Method	17110	4.709905
10	File	15202	4.184686
11	Parameter	13293	3.659191
12	Commit	13149	3.619552
13	Rename	10854	2.987803
14	ParameterizedType	7279	2.003705
15	Value	5428	1.494177
16	Type	3715	1.022636
17	Xml	3675	1.011625
18	Field	3604	0.992080
19	Annotation	2975	0.818934
20	WildcardType	2947	0.811227
21	Element	2162	0.595138
22	Constructor	2113	0.581650
23	Json	1570	0.432177
24	Text	1450	0.399144
25	Class	1327	0.365286
26	TypeVariable	1111	0.305827
27	Lambda	968	0.266463
28	ResolvedDuplicateType	910	0.250498
29	GenericDeclaration	904	0.248846
30	JavaType	754	0.207555
31	Key	702	0.193241
32	Scalar	685	0.188561
33	Primitive	672	0.184983
34	Person	551	0.151675
35	ExternalType	527	0.145068
36	Maven	346	0.095244
37	Copy	342	0.094143
38	Author	299	0.082306
39	Array	293	0.080655

Chart 1c - Highest node count by label

Shows the 40 labels with the highest number of nodes.

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Node count by label



Relationship Types

Table 2a - Highest relationship count by type

Lists the 30 relationship types with the highest number of occurrences. The whole table can be found in the CSV report `Relationship_type_count`.

Total number of relationships: 1127591

	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	CONTAINS_CHANGE	271255	24.056152
1	MODIFIES	271255	24.056152
2	UPDATES	178996	15.874196
3	CREATES	64255	5.698431
4	DELETES	38858	3.446108
5	INVOKES	36425	3.230338
6	COMMITTED	26298	2.332229
7	DEPENDS_ON	22492	1.994695
8	OF_TYPE	21920	1.943967
9	DECLARES	21175	1.877897
10	OF_RAW_TYPE	17353	1.538945
11	HAS_PARENT	15955	1.414963
12	HAS	14429	1.279631
13	HAS_COMMIT	13149	1.166114
14	RETURNS	12844	1.139065
15	HAS_FILE	11241	0.996904
16	RENAMES	10854	0.962583
17	READS	9379	0.831773
18	HAS_ACTUAL_TYPE_ARGUMENT	8407	0.745572
19	HAS_NEW_NAME	6344	0.562615
20	OF_GENERIC_TYPE	5996	0.531753
21	RESOLVES_TO	5235	0.464264
22	SIMILAR	4078	0.361656
23	WRITES	3944	0.349772
24	CONTAINS	3936	0.349063
25	RETURNS_GENERIC	3593	0.318644
26	ANNOTATED_BY	2919	0.258870
27	REQUIRES	2230	0.197767
28	HAS_FIRST_CHILD	2162	0.191736
29	HAS_LAST_CHILD	2162	0.191736

Chart 2a - Highest relationship count by type

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in the second chart.

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Relationship types (more than 0.5% overall)

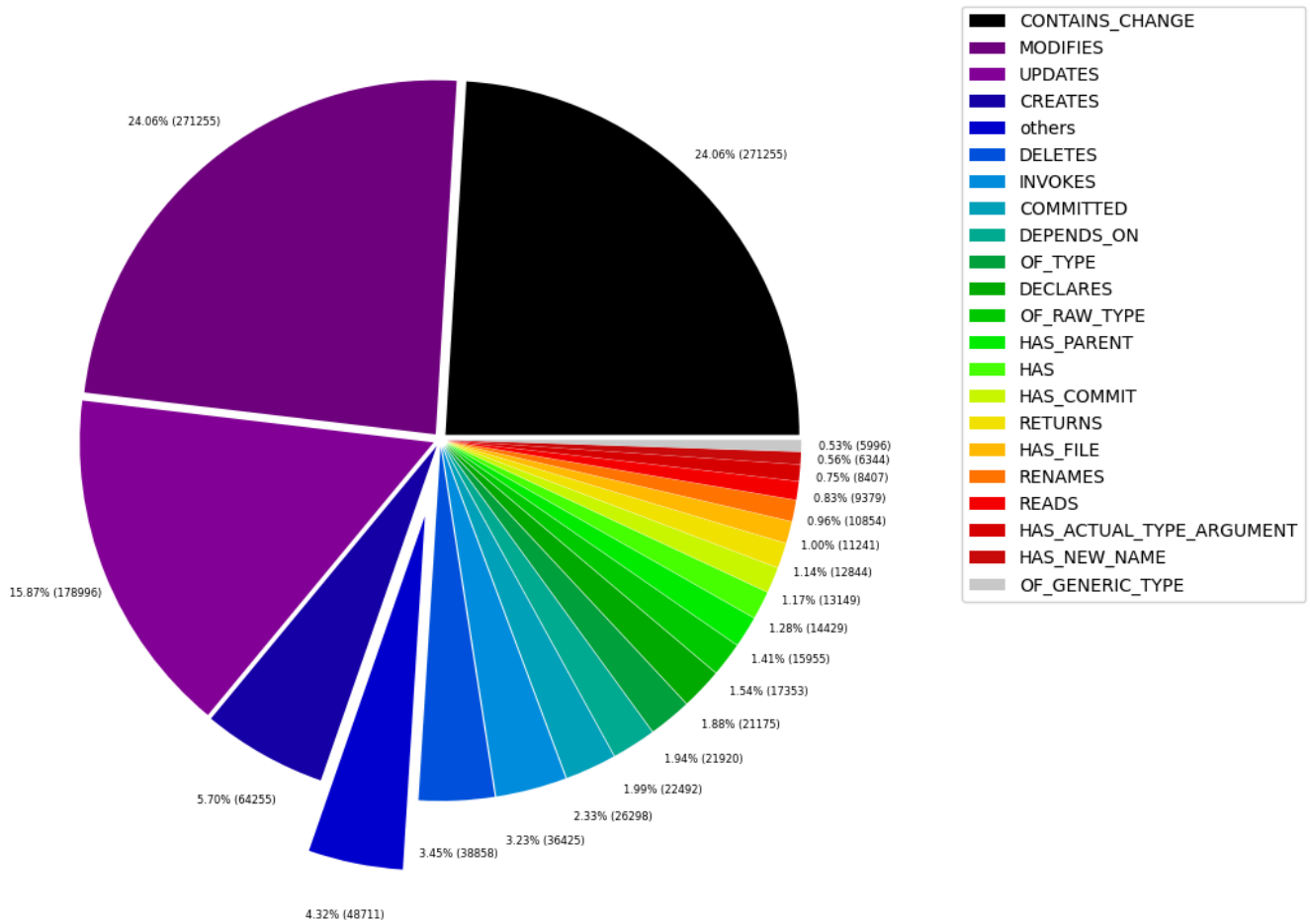


Table 2b - Lowest relationship count by type

Lists the 30 relationships type with the lowest number of occurrences up to 0.5% of the total node count. This is essentially breaking down the "others" slice from the chart above.

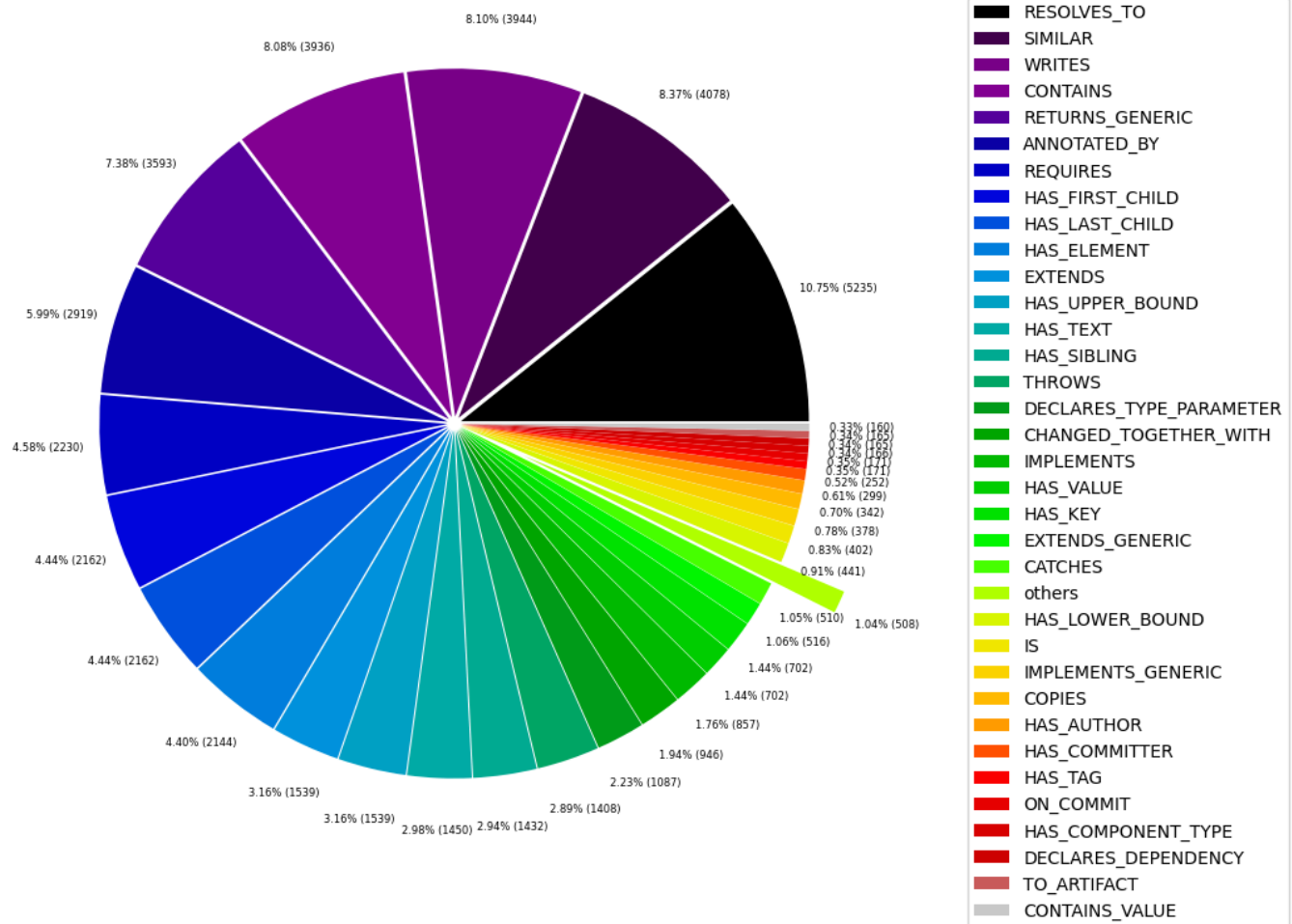
	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	HAS_PROPERTY	1	0.000089
1	THROWS_GENERIC	5	0.000443
2	EXCLUDES	5	0.000443
3	DECLARES_DEV_DEPENDENCY	7	0.000621
4	DESCRIBES	9	0.000798
5	HAS_ROOT_ELEMENT	11	0.000976
6	HAS_GOAL	16	0.001419
7	HAS_EXECUTION	16	0.001419
8	OF_NAMESPACE	18	0.001596
9	HAS_ATTRIBUTE	18	0.001596
10	INCLUDES_CONCEPT	19	0.001685
11	IS_ARTIFACT	21	0.001862
12	HAS_CONFIGURATION	21	0.001862
13	USES_PLUGIN	21	0.001862
14	REQUIRES_TYPE_PARAMETER	24	0.002128
15	REQUIRES_CONCEPT	28	0.002483
16	HAS_BRANCH	30	0.002661
17	HAS_HEAD	31	0.002749
18	DECLARES_NAMESPACE	36	0.003193
19	HAS_DEFAULT	39	0.003459
20	COPY_OF	132	0.011706
21	CONTAINS_VALUE	160	0.014190
22	DECLARES_DEPENDENCY	165	0.014633
23	TO_ARTIFACT	165	0.014633
24	HAS_COMPONENT_TYPE	166	0.014722
25	ON_COMMIT	171	0.015165
26	HAS_TAG	171	0.015165
27	HAS_COMMITTER	252	0.022349
28	HAS_AUTHOR	299	0.026517
29	COPIES	342	0.030330

Chart 2b - Lowest relationship count by type

Shows the lowest (less than 0.5% overall) relationship types. This plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

<Figure size 640x480 with 0 Axes>

Relationship types (less than 0.5% overall)



Node labels with their relationships

Table 3a - Highest relationship count by node labels and relationship type

Lists the 30 node labels and their relationship types with the highest number of occurrences.

	sourceLabels	relationType	targetLabels	numberOfRelationships	numberOfNodesWithSameLabelsAsSource	numbe
0	[Git, Update, Change]	UPDATES	[File, Git]	178996	178996	
1	[Git, Update, Change]	MODIFIES	[File, Git]	178996	178996	
2	[Git, Commit]	CONTAINS_CHANGE	[Git, Update, Change]	178996	13149	
3	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Create]	53059	13149	
4	[Git, Change, Create]	MODIFIES	[File, Git]	53059	53059	
5	[Git, Change, Create]	CREATES	[File, Git]	53059	53059	
6	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Delete]	28004	13149	
7	[Git, Change, Delete]	MODIFIES	[File, Git]	28004	28004	
8	[Git, Change, Delete]	DELETES	[File, Git]	28004	28004	
9	[Java, ByteCode, Member, Method]	INVOKES	[Java, ByteCode, Member, Method]	22175	13349	
10	[Git, Commit]	HAS_PARENT	[Git, Commit]	15946	13149	
11	[Repository, File, Git]	HAS_COMMIT	[Git, Commit]	13149	1	
12	[Committer, Git, Person]	COMMITTED	[Git, Commit]	13149	252	
13	[Author, Git, Person]	COMMITTED	[Git, Commit]	13149	299	
14	[Repository, File, Git]	HAS_FILE	[File, Git]	11241	1	
15	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Rename]	10854	13149	
16	[Git, Change, Rename]	DELETES	[File, Git]	10854	10854	
17	[Git, Change, Rename]	RENAMES	[File, Git]	10854	10854	
18	[Git, Change, Rename]	CREATES	[File, Git]	10854	10854	
19	[Git, Change, Rename]	MODIFIES	[File, Git]	10854	10854	
20	[Java, ByteCode, Member, Method]	HAS	[Java, ByteCode, Parameter]	8500	13349	
21	[Java, ByteCode, Member, Method]	READS	[Java, ByteCode, Member, Field]	8444	13349	
22	[File, Git]	HAS_NEW_NAME	[File, Git]	6344	11241	
23	[Java, ByteCode, Parameter]	OF_TYPE	[Type, File, Java, ByteCode, JavaType]	6156	13293	
24	[Java, ByteCode, Bound]	OF_RAW_TYPE	[Type, File, Java, ByteCode, JavaType]	3465	7240	
25	[Java, ByteCode, ParameterizedType, Bound]	OF_RAW_TYPE	[Type, File, Java, ByteCode, JavaType]	3110	7279	
26	[Java, ByteCode, ParameterizedType, Bound]	HAS_ACTUAL_TYPE_ARGUMENT	[Java, ByteCode, Bound, WildcardType]	2947	7279	
27	[Java, ByteCode, Parameter]	OF_GENERIC_TYPE	[Java, ByteCode, ParameterizedType, Bound]	2695	13293	
28	[Java, ByteCode, Member, Method, Constructor]	WRITES	[Java, ByteCode, Member, Field]	2519	2109	
29	[Java, ByteCode, ParameterizedType, Bound]	HAS_ACTUAL_TYPE_ARGUMENT	[Java, ByteCode, Bound, TypeVariable]	2474	7279	

Graph Density

```
total_number_of_nodes (vertices): 363277
total_number_of_relationships (edges): 1127591
-> total directed graph density: 8.544309894940536e-06
-> total directed graph density in percent: 0.0008544309894940536
```